

AVISTA Model Report

An extensible desktop platform for tabular machine learning and deep learning analytics.

Project: mushroom

Target: poisonous

Dataset rows: 8124

Features: 13

Split rows: 5686 train / 813 validation / 1625 test

Imbalance method: smote

Validation role: deep-model early stopping and checkpoint selection

Test role: held-out final evaluation after fitting and checkpoint selection

CV role: outer validation folds are used only for fold scoring

Cross-validation: 5 folds

Models trained: 3

Generated on: 2026-09-19T18:58:07

AVISTA version: 1.0.7

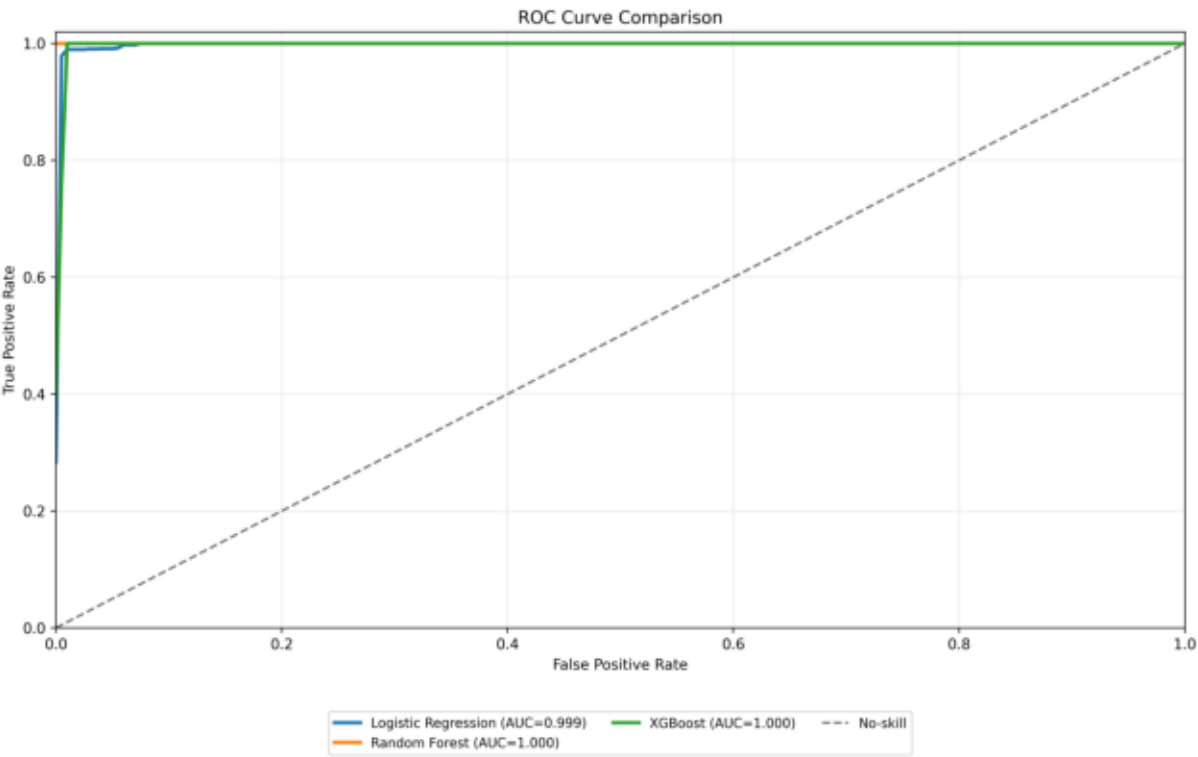
AVISTA release date: July 28, 2026

Unless otherwise stated, model comparison figures and diagnostic tables are based on the test set.

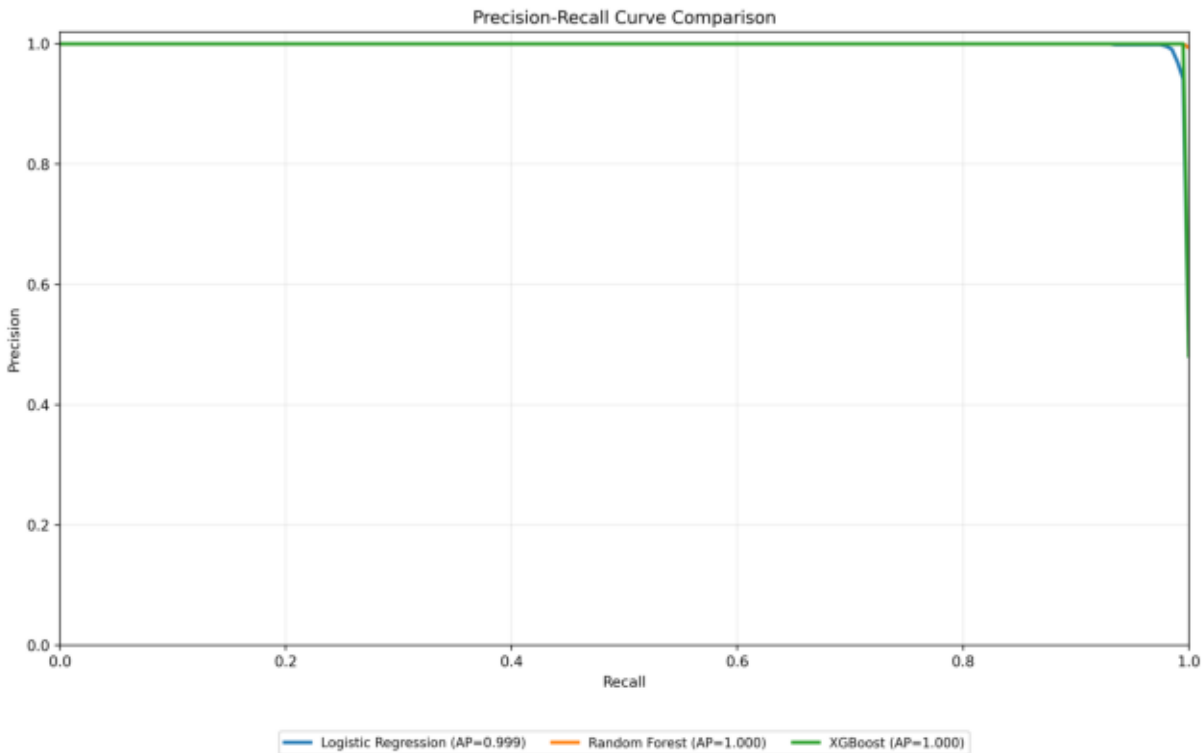
Model Performance Summary

Model	Status	Validation Accuracy	Validation Macro-F1	Test Accuracy	Test Macro-F1	ROC-AUC	Saved
Logistic Regression	trained	0.9865	0.9864	0.9760	0.9760	0.9993	Yes
Random Forest	trained	1.0000	1.0000	1.0000	1.0000	1.0000	Yes
XGBoost	trained	1.0000	1.0000	1.0000	1.0000	1.0000	Yes

ROC Curve Comparison

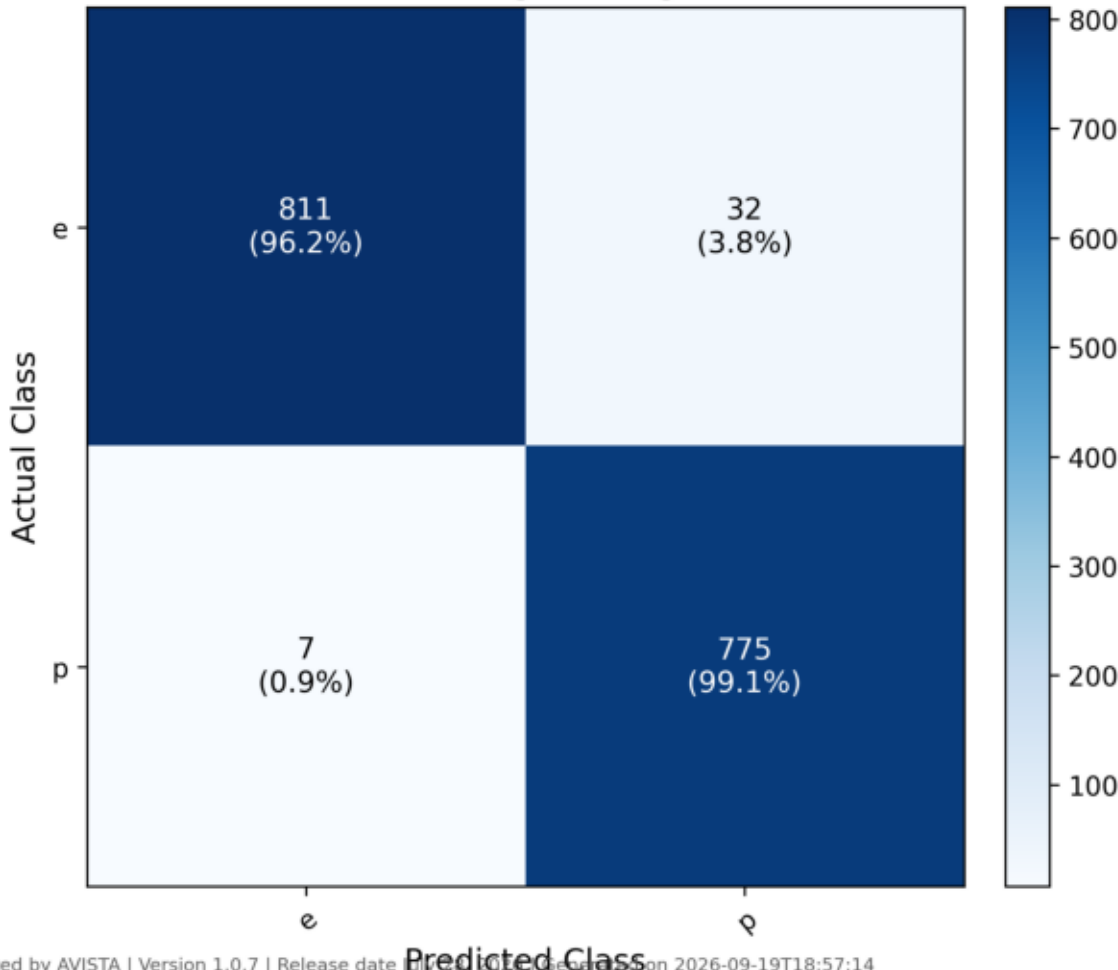


Precision-Recall Curve Comparison



Confusion Matrices - Logistic Regression

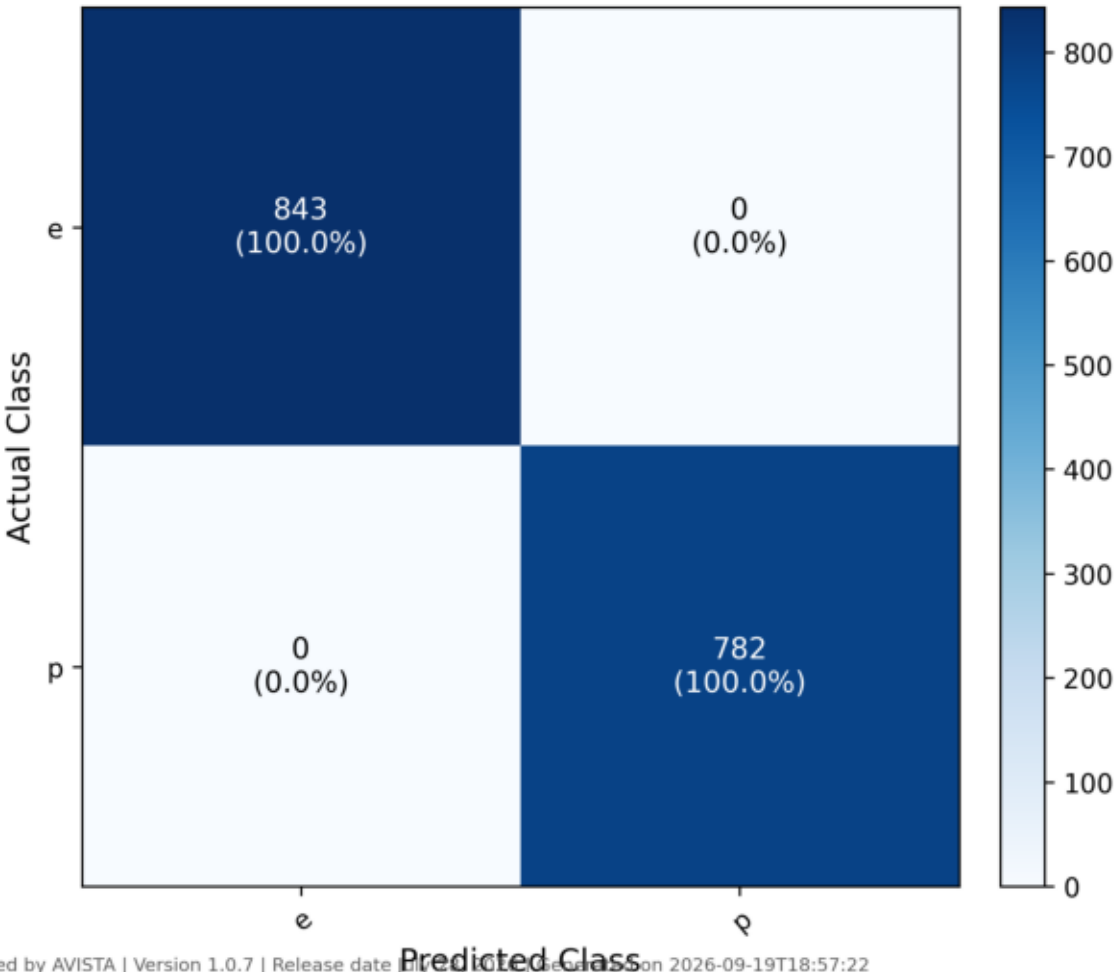
Confusion Matrix - Logistic Regression - Test



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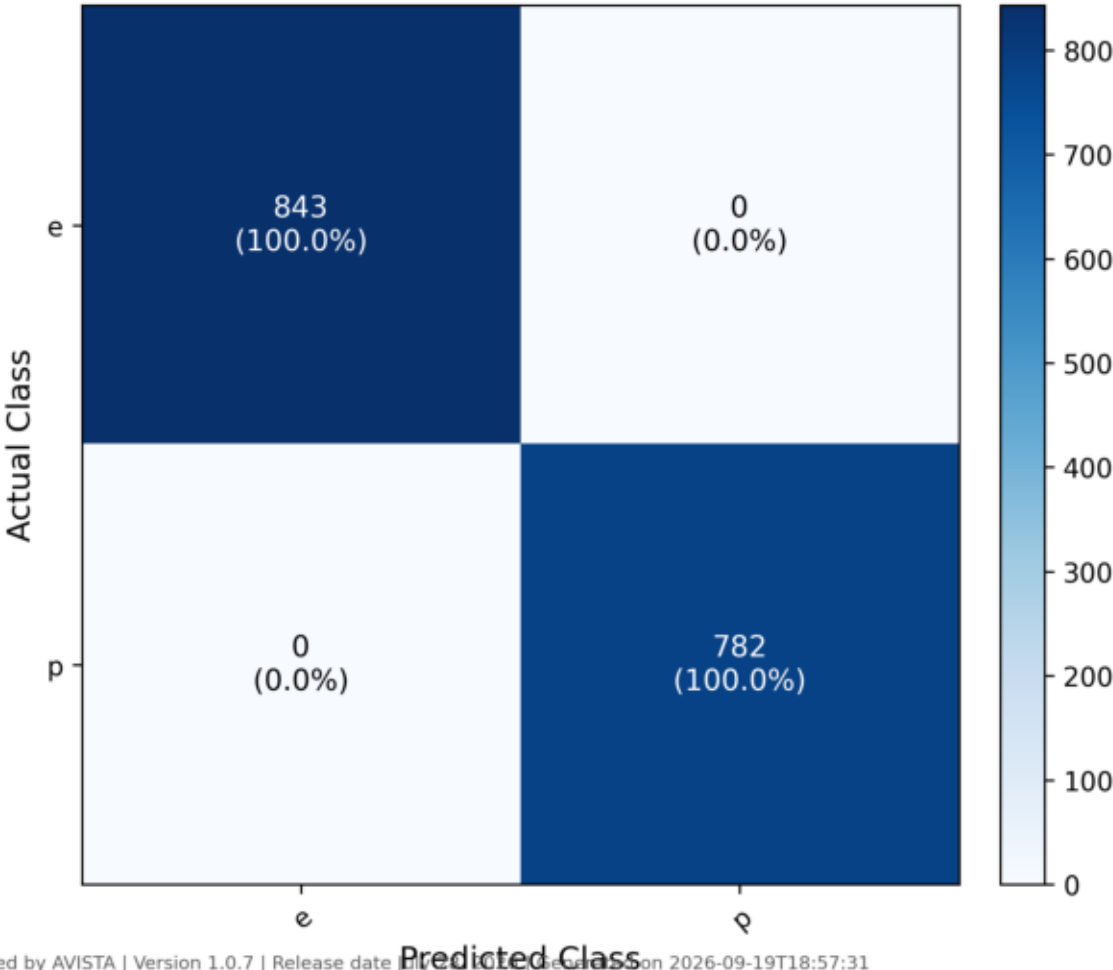
Confusion Matrices - Random Forest

Confusion Matrix - Random Forest - Test

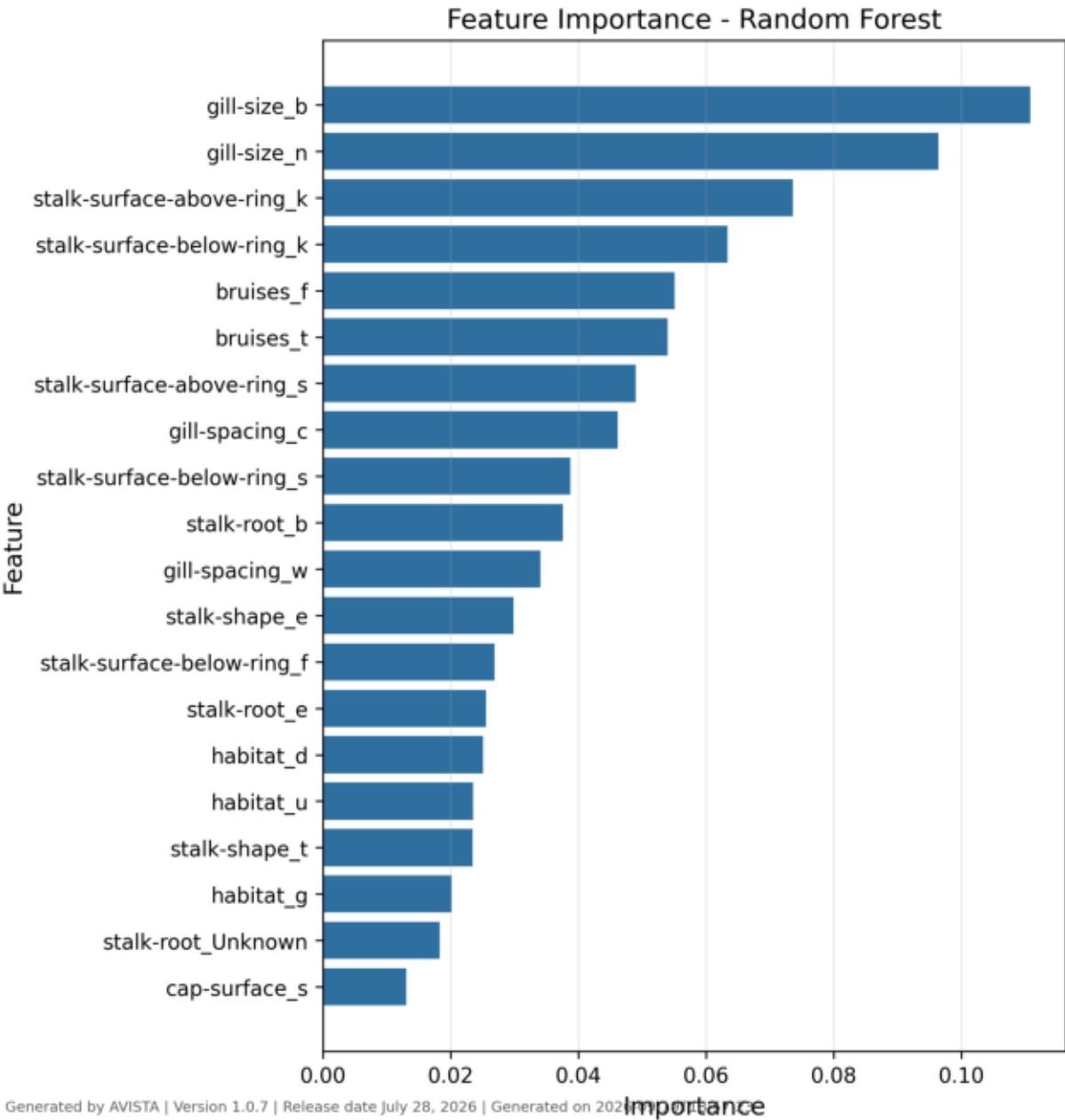


Confusion Matrices - XGBoost

Confusion Matrix - XGBoost - Test



Feature Importance - Random Forest



Test Classification Report - Logistic Regression

Class	precision	recall	f1-score	support
e	0.9914	0.9620	0.9765	843.0000
p	0.9603	0.9910	0.9755	782.0000
accuracy	0.9760	0.9760	0.9760	0.9760
macro avg	0.9759	0.9765	0.9760	1625.0000
weighted avg	0.9765	0.9760	0.9760	1625.0000

Test Classification Report - Random Forest

Class	precision	recall	f1-score	support
e	1.0000	1.0000	1.0000	843.0000
p	1.0000	1.0000	1.0000	782.0000
accuracy	1.0000	1.0000	1.0000	1.0000
macro avg	1.0000	1.0000	1.0000	1625.0000
weighted avg	1.0000	1.0000	1.0000	1625.0000

Test Classification Report - XGBoost

Class	precision	recall	f1-score	support
e	1.0000	1.0000	1.0000	843.0000
p	1.0000	1.0000	1.0000	782.0000
accuracy	1.0000	1.0000	1.0000	1.0000
macro avg	1.0000	1.0000	1.0000	1625.0000
weighted avg	1.0000	1.0000	1.0000	1625.0000